

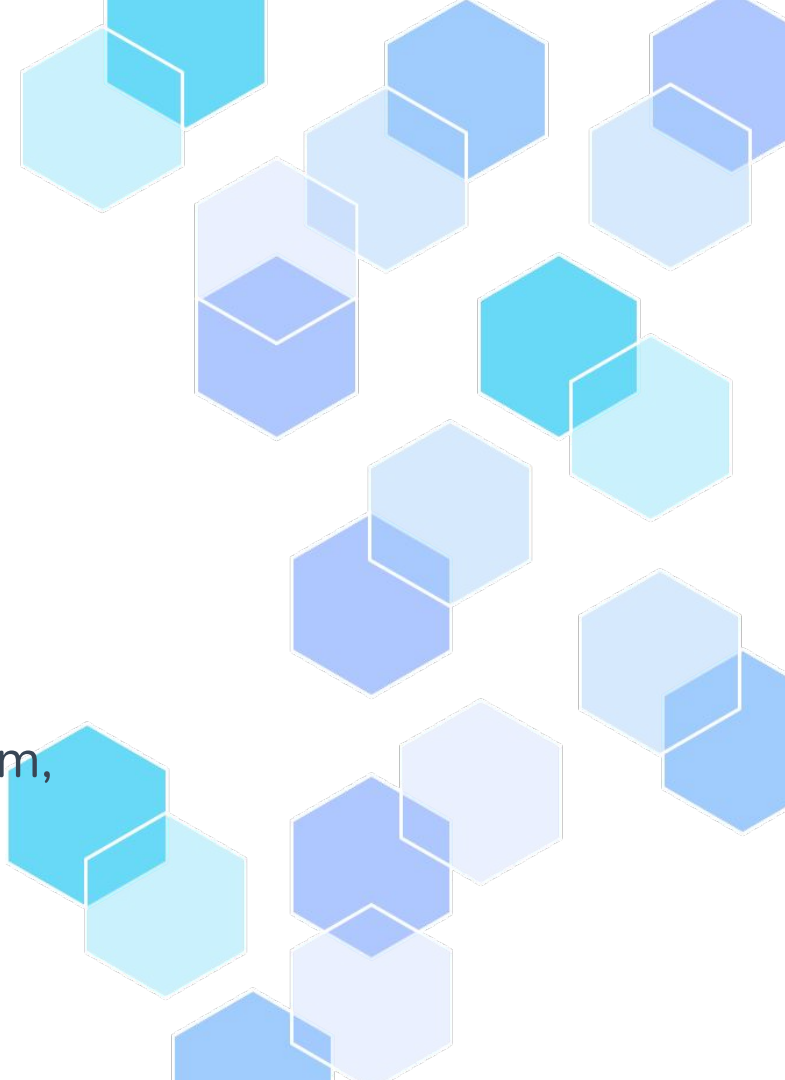
Deepfake Generation Method Classification Using Artifact-Oriented Features

Offensive AI

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Introduction

Deepfake technology has reached near-photorealistic quality. Most existing research focuses on Binary Classification (Real vs. Fake).
Limitation: Binary detection offers limited forensic value.

The Proposed Solution:

Multi-class Attribution– Identifying which specific manipulation method was used.

Enables attribution to tools, characterizes capabilities, and supports targeted defense.

Manipulation Pipelines

- **DeepFakes:** Autoencoder-based identity swap
- **Face2Face:** Expression reenactment transfer
- **FaceSwap:** Classical 3D face alignment and texture transfer
- **NeuralTextures:** Neural rendering-based texture synthesis

Artifact-Oriented Features

1. FFT (Radial)

Spectral energy distribution across low, mid, and high (frame-wise)

- frequency radial bands captures **upsampling noise** artifacts inserted by **generators models** when frame created

2. Geometric Jitter

Temporal **facial edge centroids displacement**

- measures geometric instability in facial structure **across frames**.

3. Pixel Temporal

Frame-to-frame intensity differences and high-frequency spectral energy

- measures **pixel-level values** inconsistencies across consecutive frames

Artifact-Oriented Features

4. Face Embeddings

Temporal descriptors from pre-trained model **InceptionResnetV1**

- Face identity into **deep feature vector**
- Captures **identity** and **texture-level**(e.g. skin details) variations across frames.

5. Background

Temporal embeddings (frames-series) of **background** regions

- analyzes manipulation artifacts outside the face area (blurred background)

6. Optical Flow

Pixel-wise warping **residuals** between consecutive frames

- quantifies temporal incoherence and inconsistency from independent frame synthesis.

Datasets & Preprocessing

Dataset: FaceForensics++ (c23 compression)

- Selection: 120 videos per method (480 total)
- Constraint: Only manipulated videos used (focus on attribution).

Preprocessing:

- Sampling: 24 frames uniformly sampled per video.
- Face Detection: MTCNN used to crop faces (160x160 pixels)
- Filtering: Videos with <18 valid detections excluded.

Methodology Overview

- **Pipeline:**
 - **Input:** Videos from FaceForensics++.
 - **Preprocessing:** Frame sampling & Face Detection (MTCNN)
 - **Feature Extraction:** Artifact-oriented signals (Frequency, Geometry, Temporal).
 - **Classification**

Experimental Setup

- **Classifiers:**
 - Random Forest: 300 trees, balanced weights.
 - SVM
 - XGBoost
- **Evaluation Tasks:**
 - One-vs-All (Binary per method).
 - Pairwise (Discriminating 6 unique pairs).
 - Multi-class (4-way classification).
- **Validation:**
 - Stratified 10-fold cross-validation.

Results

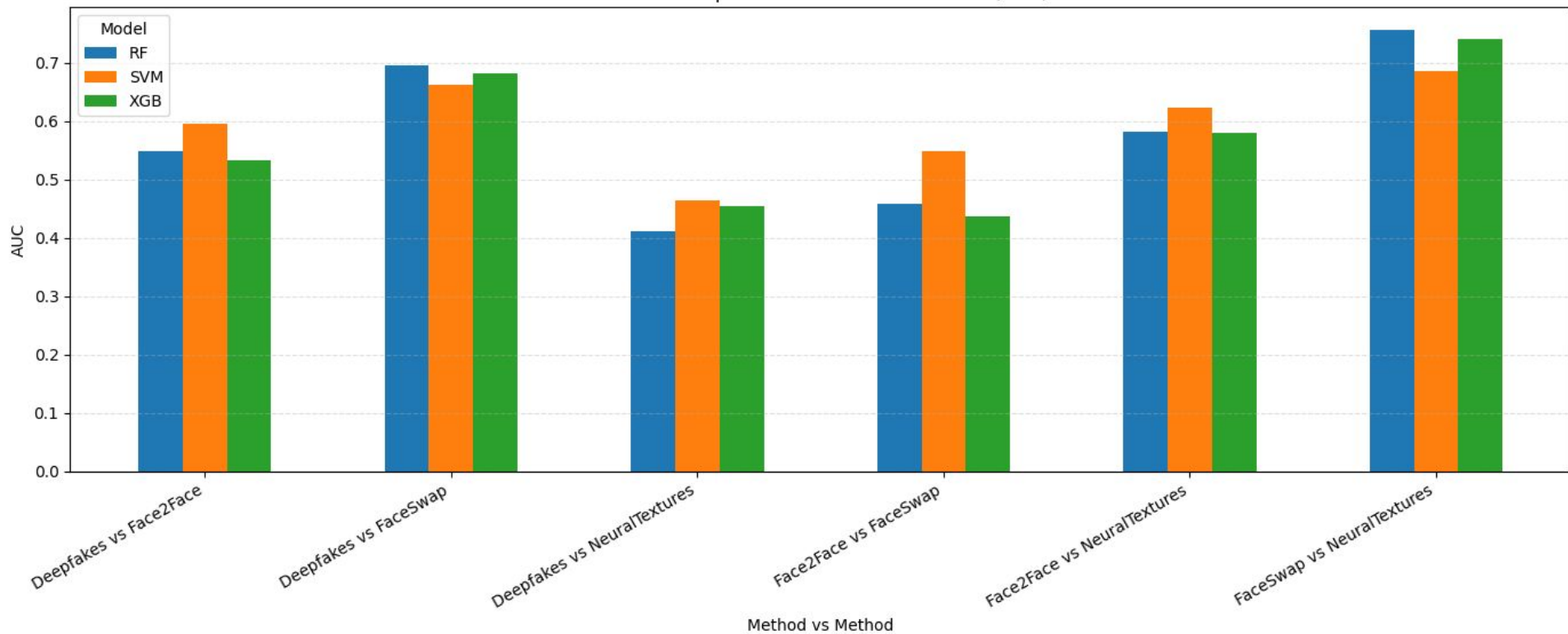
Multi Class problem

Feature Type	Accuracy (%)	AUC (%)
FFT (Radial)	26.0 ± 6.3	54.3 ± 4.3
Landmark Jitter	24.6 ± 5.7	47.6 ± 6.6
Optical Flow	25.6 ± 5.7	49.7 ± 4.8
Face Embeddings	38.8 ± 6.2	66.9 ± 4.5
Background	20.8 ± 4.6	39.9 ± 3.1
Pixel Temporal	24.0 ± 4.9	47.8 ± 5.3
Random Baseline	25.0	50.0

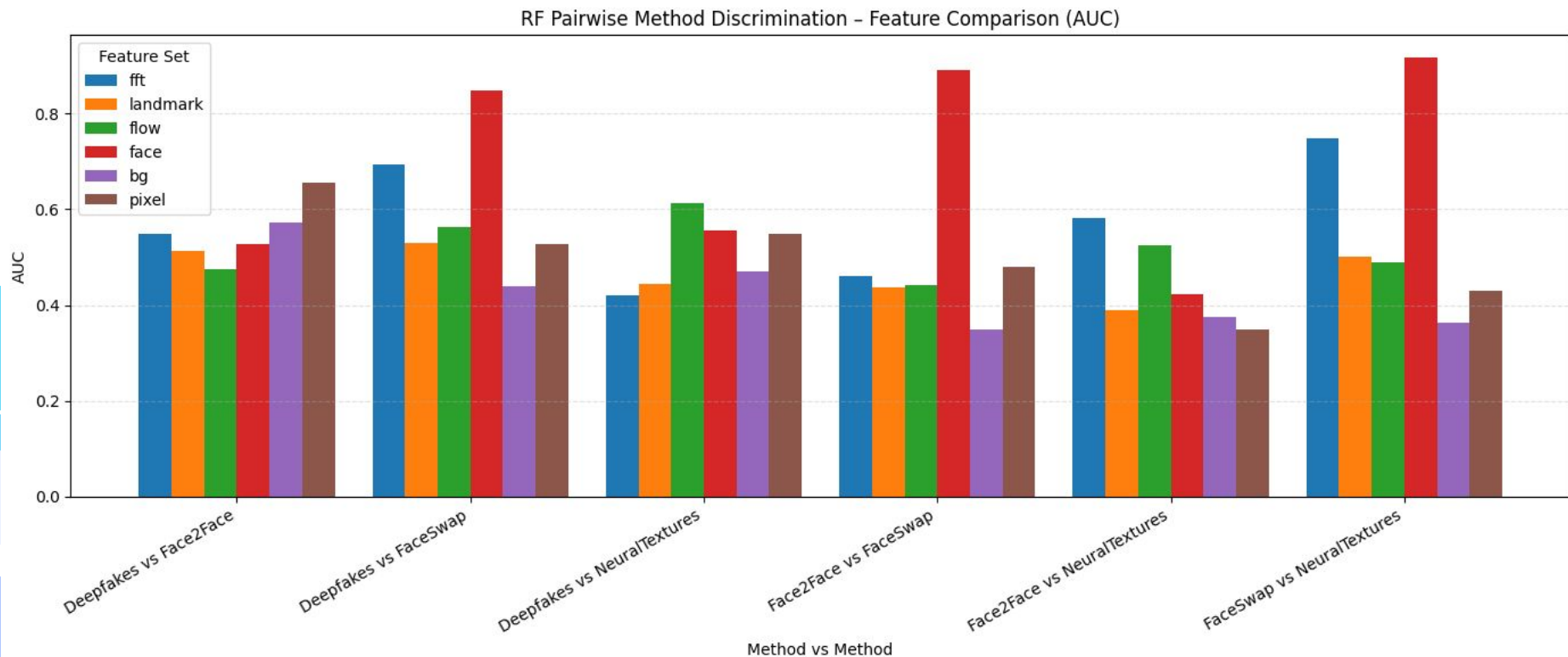
- Random Forest classifier
- Best Performance is Face Embeddings 38.8% Acc, 66.9% AUC
- Second performer is FFT (Radial): 26.0% Acc, 54.3% AUC.
- Optical Flow / Landmark Jitter: Near chance (47–50% AUC).

Pairwise Binary Classification - Model selection

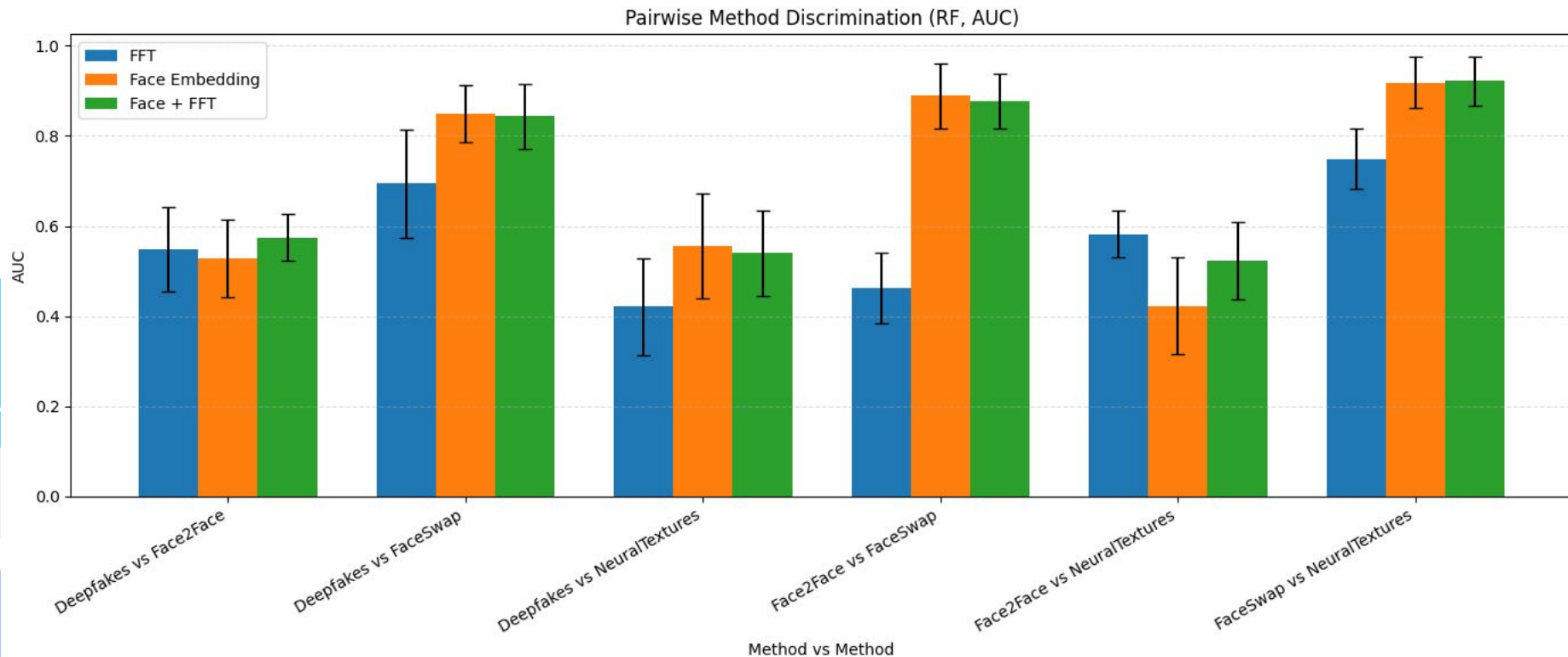
Pairwise Deepfake Method Discrimination (AUC)



Pairwise Binary Classification - Feature Selection



Pairwise Binary Classification - Feature Combination

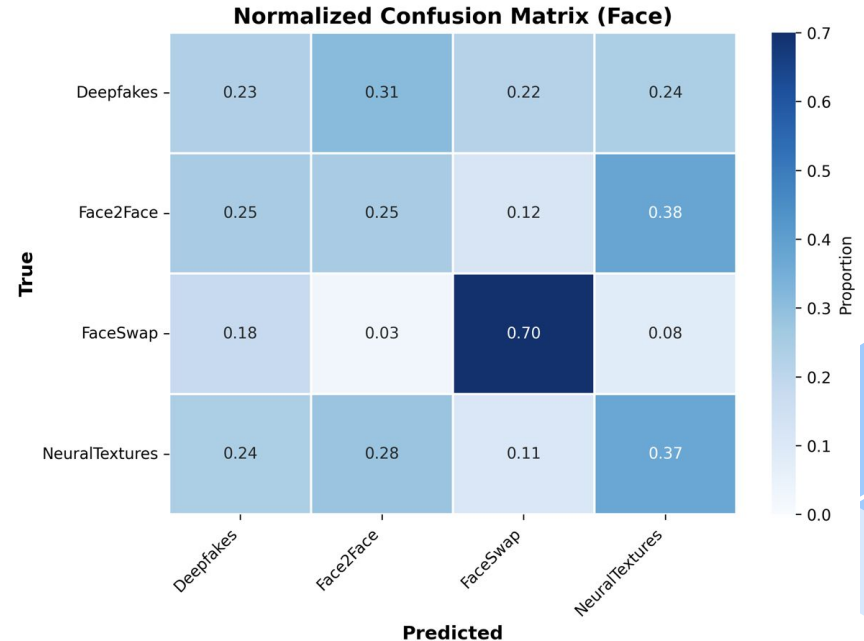


Back to Multi Class problem

Error Analysis

Observations from Face Confusion Matrix:

- FaceSwap is easier to detect using the Face embedding feature
- Deepfakes signatures are harder to detect using the Face feature



Back to Multi Class problem

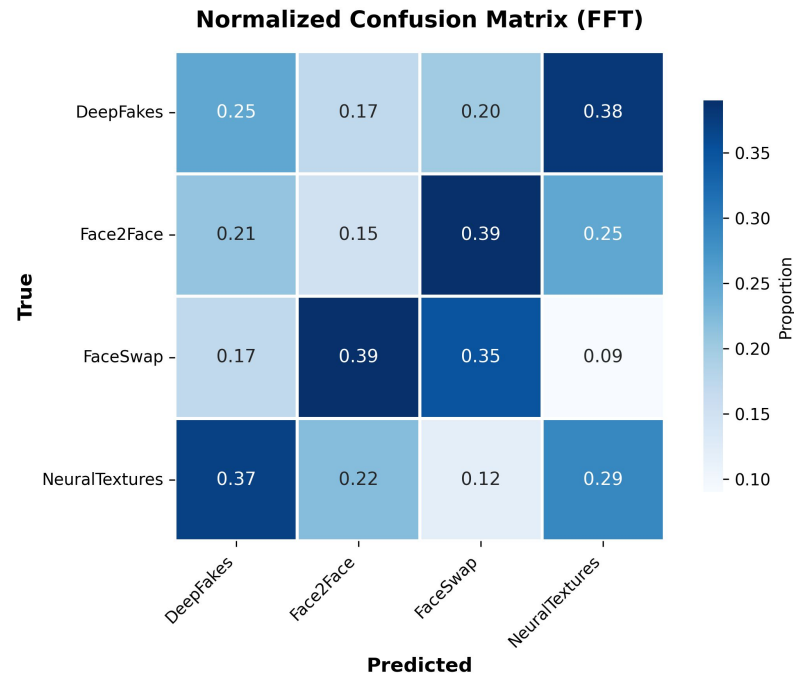
Error Analysis

Observations from FFT Confusion Matrix:

- DeepFakes confused with NeuralTextures (38%)
 - Both use neural synthesis.
- Face2Face confused with FaceSwap (39%)
 - Both rely on 3D geometry

Conclusion:

- Methods sharing underlying technical paradigms produce overlapping artifact distributions.



Conclusions

- Frequency-domain cues are critical for neural rendering methods
- Identity embeddings dominate classical swap methods
- Temporal instability provides complementary information
- Feature fusion must be applied selectively

Questions?
Thank you!